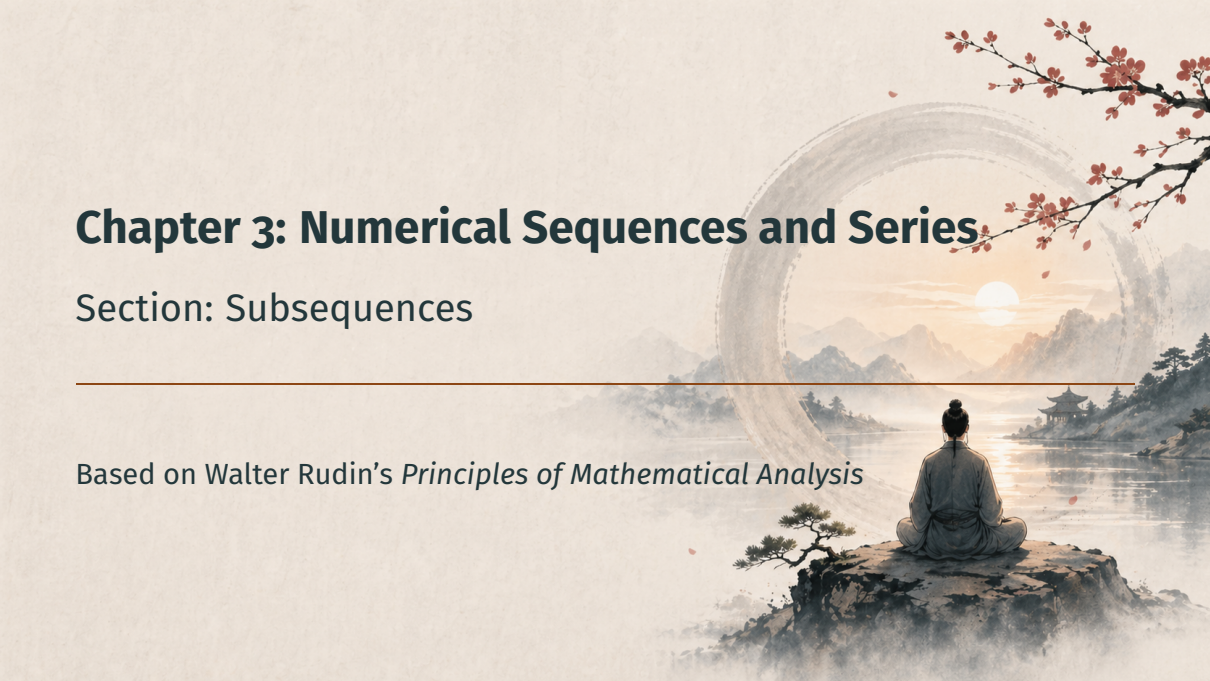


Chapter 3: Numerical Sequences and Series

The background is a traditional Chinese ink wash painting. A person with their hair in a bun, wearing a grey robe, sits in a meditative lotus position on a dark, craggy rock in the foreground. They are looking out over a calm lake that reflects the sky. In the distance, misty mountains rise. A large, faint circular frame, resembling a ring or a portal, is centered in the background, framing a bright sun or moon. A small pavilion with a traditional roof is visible on the right side of the lake. Red cherry blossom branches with small flowers hang down from the top right corner. A thin horizontal line is drawn across the middle of the image, separating the title from the subtitle.

Section: Subsequences

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

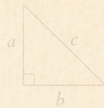
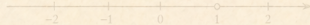
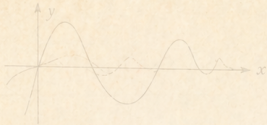
Motivation

Definition of a Subsequence

Existence of Convergent Subsequences

Subsequential Limits Form a Closed Set

Summary

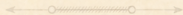


$$a^2 + b^2 = c^2$$

Motivation



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Why Subsequences?

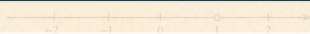
- Many sequences **do not converge** ... but still have structure



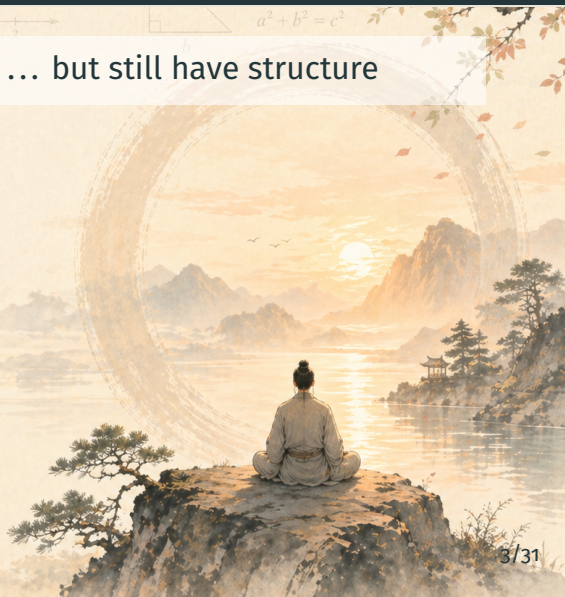
$$f(x) = \sin x$$



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Why Subsequences?

- Many sequences **do not converge** ... but still have structure
- A **subsequence** extracts a thinned-out piece that may converge



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Why Subsequences?

- Many sequences **do not converge** ... but still have structure
- A **subsequence** extracts a thinned-out piece that may converge
- In **compact** spaces, a convergent subsequence *always exists*

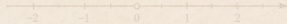


$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Definition of a Subsequence



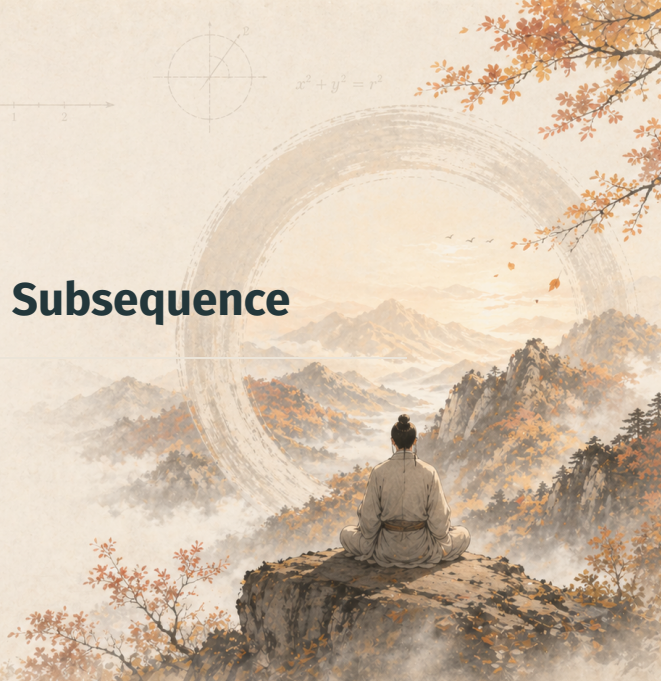
$$x^2 + y^2 = r^2$$



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



Definition 3.5: Subsequence

Definition 3.5 — Subsequence

Given a sequence $\{p_n\}$ and a sequence of **positive integers**

$$n_1 < n_2 < n_3 < \cdots ,$$

the sequence $\{p_{n_k}\}$ is called a **subsequence** of $\{p_n\}$.



$$a^2 + b^2 = c^2$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

Definition 3.5: Subsequential Limit

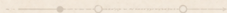
Definition 3.5 — Subsequential Limit

Given $\{p_n\}$ and a subsequence $\{p_{n_k}\}$:

- If $\{p_{n_k}\}$ converges, its limit is a **subsequential limit** of $\{p_n\}$.
- Write E^* for the set of all subsequential limits.



$$a^2 + b^2 = c^2$$



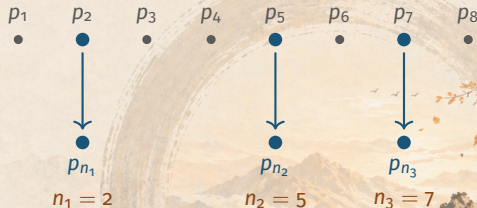
$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



Picture: A Subsequence Is a Thinned Sequence

Original $\{p_n\}$:

Subsequence $\{p_{n_k}\}$:



Keep some terms (in order), drop the rest.



$$a^2 + b^2 = c^2$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

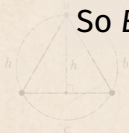
Example: $p_n = (-1)^n$

Two Subsequential Limits

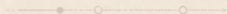
Let $p_n = (-1)^n$ in $\mathbb{R}$. The sequence $\{p_n\}$ **diverges**, but:

- Even indices ($n_k = 2k$): $p_{n_k} = 1 \rightarrow 1$.
- Odd indices ($n_k = 2k - 1$): $p_{n_k} = -1 \rightarrow -1$.

So $E^* = \{-1, +1\}$.



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

Remark: Convergence and Subsequences

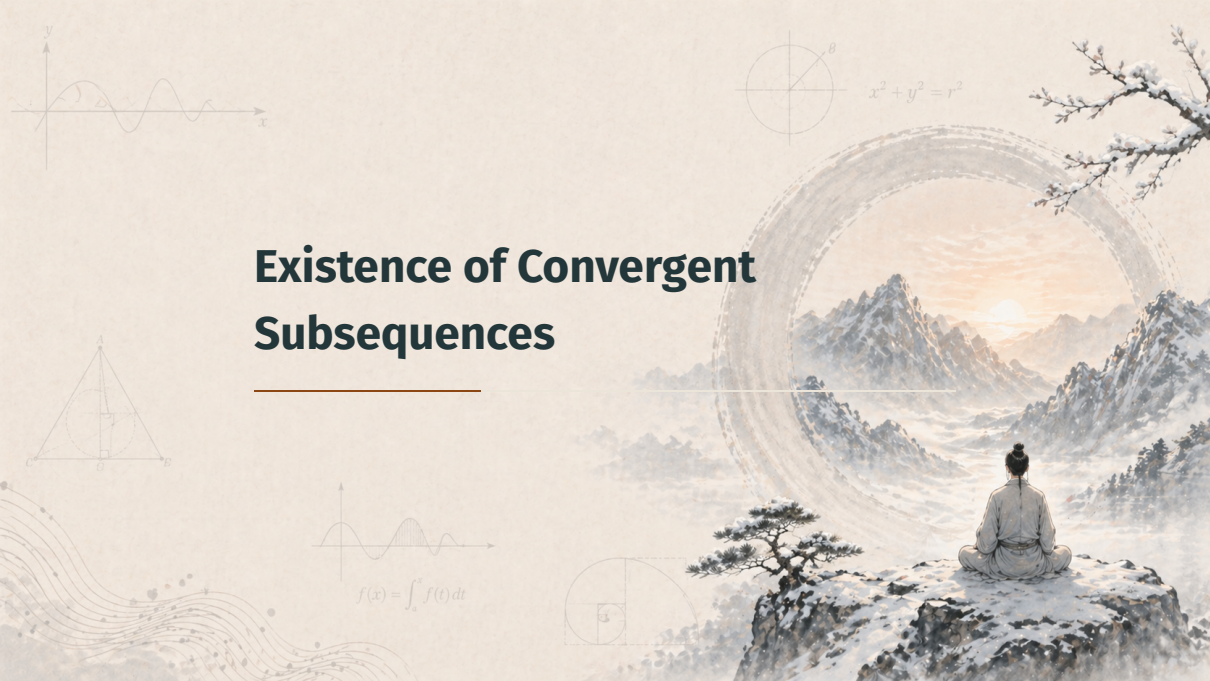
Useful Equivalence

$p_n \rightarrow p \iff$ every subsequence $\{p_{n_k}\}$ converges to p .

Two-way picture:

- $(\Rightarrow)$ Tail of $\{p_n\}$ in ε -ball $\implies$ tail of $\{p_{n_k}\}$ in ε -ball.
- $(\Leftarrow)$ The original $\{p_n\}$ is *itself* a subsequence.

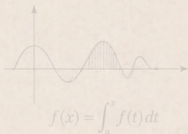
Existence of Convergent Subsequences



Theorem 3.6: Compactness Forces Convergence

Theorem 3.6 (a)

If $\{p_n\}$ is a sequence in a **compact** metric space X , then some subsequence of $\{p_n\}$ converges to a point of X .



Theorem 3.6: Compactness Forces Convergence

Theorem 3.6 (a)

If $\{p_n\}$ is in a **compact** metric space X , then some subsequence converges in X .

Theorem 3.6 (b)

Every **bounded** sequence in $\mathbb{R}^k$ contains a convergent subsequence.

Proof Strategy: Theorem 3.6 (a)

Goal: Build a convergent subsequence of $\{p_n\}$ inside compact X .

Two cases by the range $E = \{p_n : n \geq 1\}$:

- (I) E is **finite**: some value p is hit infinitely often $\Rightarrow$ constant subsequence.
- (II) E is **infinite**: by Theorem 2.37, E has a limit point $p \in X$.
Build a subsequence inductively, using $\varepsilon = 1/k$ at step k .

Proof of 3.6 (a) — Case I: E Finite

Case I: finite range

If $E = \{p_n : n \geq 1\}$ is finite, some value $p \in E$ is taken **infinitely often**:

$$p_{n_1} = p_{n_2} = p_{n_3} = \cdots = p, \quad n_1 < n_2 < n_3 < \cdots$$

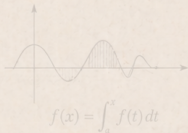
The constant subsequence $\{p_{n_k}\}$ converges to $p \in X$. $\square$

Proof of 3.6 (a) — Case II: Step 1

Case II: E infinite, Step 1 — find a limit point

By **Theorem 2.37**, every infinite subset of a compact set has a limit point in the set.

$\implies E$ has a limit point $p \in X$.



Proof of 3.6 (a) — Case II: Step 2

Step 2 — pick the first index n_1

Since p is a limit point of E , every neighborhood of p contains points of E other than p .

Choose n_1 with

$$d(p, p_{n_1}) < 1.$$

Proof of 3.6 (a) — Case II: Step 3

Step 3 — inductive choice of n_k

Suppose $n_1 < n_2 < \dots < n_{k-1}$ have been chosen.

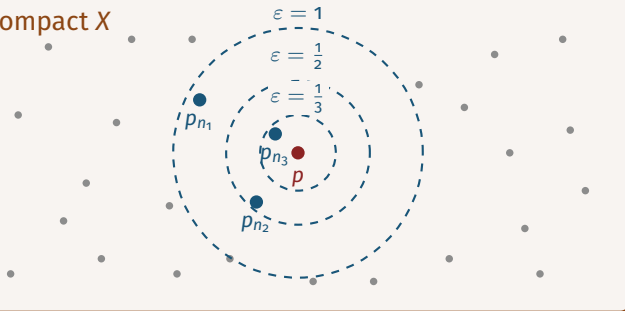
By **Theorem 2.20**: every neighborhood of a limit point contains *infinitely many* points of E .

So there is $n_k > n_{k-1}$ with

$$d(p, p_{n_k}) < \frac{1}{k}.$$

Picture: Inductive Construction of the Subsequence

compact X



Each step shrinks ε to $1/k$ and pulls one new term into the smaller ball.

$$f(x) = \int_a^x f(t) dt$$

Proof of 3.6 (a) — Case II: Step 4

Step 4 — the subsequence converges to p

For all $k \geq 1$:

$$d(p, p_{n_k}) < \frac{1}{k}.$$

Given $\varepsilon > 0$, choose $N \geq 1/\varepsilon$. Then for all $k \geq N$:

$$d(p, p_{n_k}) < \frac{1}{k} \leq \frac{1}{N} \leq \varepsilon.$$

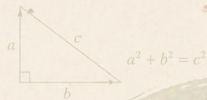
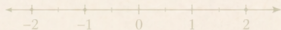
Hence $p_{n_k} \rightarrow p \in X$. $\square$

Proof of 3.6 (b) — The Bounded Case in $\mathbb{R}^k$

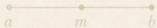
Reduction to part (a)

Let $\{p_n\}$ be a bounded sequence in $\mathbb{R}^k$.

- **Theorem 2.41:** every bounded subset of $\mathbb{R}^k$ lies in a compact set K .
- So $\{p_n\}$ is a sequence in compact K .
- Apply part (a) $\implies$ a subsequence converges in $K \subset \mathbb{R}^k$. $\square$



Subsequential Limits Form a Closed Set



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



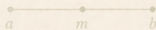
Theorem 3.7: E^* Is Closed

Theorem 3.7

The set E^* of subsequential limits of a sequence $\{p_n\}$ in a metric space X is a **closed** subset of X .

Why this matters:

- Limits of subsequential limits are again subsequential limits.
- E^* contains all its limit points — nothing escapes.

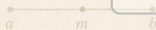
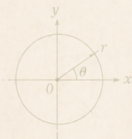
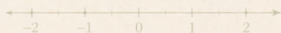

$$a \quad m \quad b$$


$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

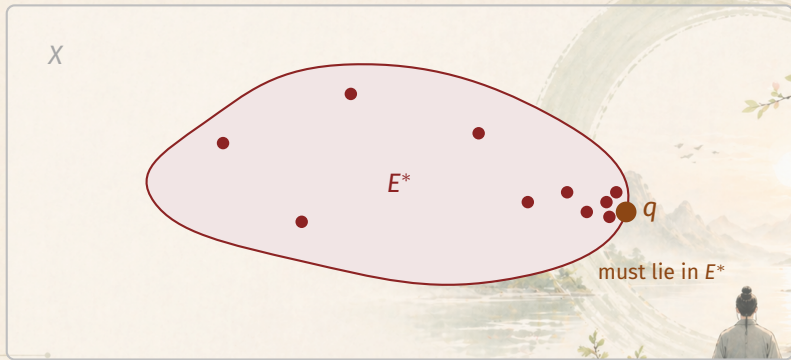
$$a^2 + b^2 = c^2$$

Picture: E^* as a Closed Set



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Proof Strategy: Theorem 3.7

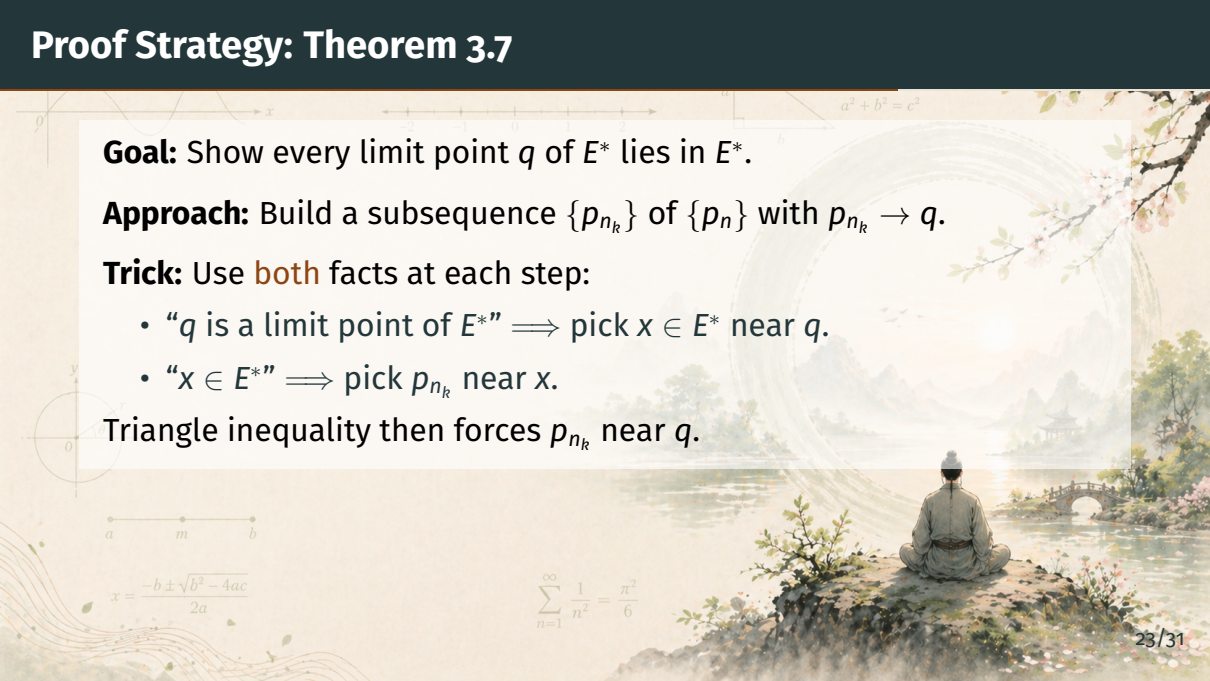
Goal: Show every limit point q of E^* lies in E^* .

Approach: Build a subsequence $\{p_{n_k}\}$ of $\{p_n\}$ with $p_{n_k} \rightarrow q$.

Trick: Use **both** facts at each step:

- “ q is a limit point of E^* ” $\implies$ pick $x \in E^*$ near q .
- “ $x \in E^*$ ” $\implies$ pick p_{n_k} near x .

Triangle inequality then forces p_{n_k} near q .



The background of the slide is a collage. At the top left, there is a coordinate system with a curve passing through the origin. Below it is a number line with points labeled -2, -1, 0, 1, 2. To the right of the number line is a right triangle with sides labeled a, b, and hypotenuse c, with the equation $a^2 + b^2 = c^2$ written above it. In the bottom left, there is a diagram of a circle with a point x on its circumference and a radius line. Below that is a diagram of a line segment with points a, m, and b. In the bottom center, there is the quadratic formula:
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
 To the right of the quadratic formula is the Basel problem sum:
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$
 On the right side of the slide is a traditional Chinese landscape painting. It shows a person sitting in a meditative pose on a rocky outcrop, looking out over a misty lake. In the distance, there are mountains and a small pavilion. A bridge is visible on the right side of the lake. The painting is in a soft, painterly style with a warm color palette.

$$a \quad m \quad b$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Proof of 3.7 — Step 1: Setup

Step 1 — the construction plan

Let q be a limit point of E^* .

Set $n_0 = 0$ (so that “ $n_k > n_{k-1}$ ” makes sense at $k = 1$).

We will build $0 < n_1 < n_2 < n_3 < \dots$ inductively so that

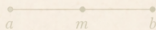
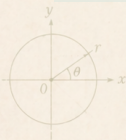
$$d(q, p_{n_k}) \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Proof of 3.7 — Step 2: Inductive Choice

Step 2 — inductive choice of n_k

Suppose $n_1, \dots, n_{k-1}$ are chosen. Pick:

- $x \in E^*$ with $d(x, q) < \frac{1}{k}$ (possible: q is a limit point of E^*).
- $n_k > n_{k-1}$ with $d(x, p_{n_k}) < \frac{1}{k}$ (possible: $x \in E^*$).



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

Proof of 3.7 — Step 3: Triangle Inequality

Step 3 — estimate $d(q, p_{n_k})$

By the triangle inequality:

$$d(q, p_{n_k}) \leq d(q, x) + d(x, p_{n_k}) < \frac{1}{k} + \frac{1}{k} = \frac{2}{k}.$$

Hence

$$d(q, p_{n_k}) < \frac{2}{k} \quad \text{for all } k \geq 1.$$

Proof of 3.7 — Step 4: Conclusion

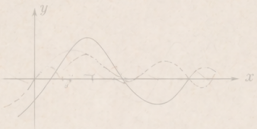
Step 4 — $\{p_{n_k}\}$ converges to q

Since $2/k \rightarrow 0$ as $k \rightarrow \infty$:

$$d(q, p_{n_k}) \rightarrow 0 \implies p_{n_k} \rightarrow q.$$

So $q \in E^*$.

Hence E^* contains all its limit points: E^* is **closed**. $\square$

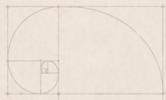


$$e^{i\theta} = \cos \theta + i \sin \theta$$



$$\nabla^2 \phi = 0$$

Summary



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



Summary

Key takeaways:

- **Definition 3.5:** subsequence $\{p_{n_k}\}$ via strictly increasing indices.

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- **Theorem 3.6:** compact $\Rightarrow$ a convergent subsequence exists; bounded in $\mathbb{R}^k \Rightarrow$ same.


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Key takeaways:

- **Definition 3.5:** subsequence $\{p_{n_k}\}$ via strictly increasing indices.
- **Convergence test:** $p_n \rightarrow p$ iff every subsequence converges to p .
- **Theorem 3.6:** compact $\Rightarrow$ a convergent subsequence exists; bounded in $\mathbb{R}^k \Rightarrow$ same.
- **Theorem 3.7:** the set E^* of subsequential limits is closed.

Coming up next: Cauchy sequences.